## Problem 2

# Problem 2

**Problem 1** *State the form of the limit. Determine whether the form is determinate or indeterminate. Evaluate each limit.*

(a) $\lim_{x\to\infty} \left(\ln(1+e^{-x})\right)^x$

**Explanation.** Since $\lim_{x\to\infty}(1+e^{-x}) = 1+0 = 1$ and $\ln(1)=0$, this limit is of the form $0^\infty$. This is a determinate form. $\lim_{x\to\infty} \left(\ln(1+e^{-x})\right)^x = 0$.

(b) $\lim_{x\to\infty} \left(\frac{1}{x}+1\right)^{\frac{1}{x}}$

**Explanation.** This limit is of the form $1^0$, which is a determinate form. Thus, $\lim_{x\to\infty} \left(\frac{1}{x}+1\right)^{\frac{1}{x}} = 1$

(c) $\lim_{x\to\infty} \left(\frac{\arctan(x)}{x}\right)$

**Explanation.** Since $\lim_{x\to\infty} \arctan(x) = \frac{\pi}{2}$, this limit is of the form $\frac{\#}{\infty}$, which is a determinate form.

Thus, $\lim_{x\to\infty} \left(\frac{\arctan(x)}{x}\right) = 0$

(d) $\lim_{x\to\infty} (x - \ln(x))$

**Explanation.** This limit is of the form $\infty - \infty$, which is an indeterminate form. We can rewrite this as:

$$\lim_{x\to\infty} (x-\ln(x)) = \lim_{x\to\infty} \left(x\left(1-\frac{\ln(x)}{x}\right)\right)$$

We can see that $\lim_{x\to\infty} \left(\frac{\ln(x)}{x}\right)$ is of the form $\frac{\infty}{\infty}$, so we can use L'Hopital's Rule.

$$\stackrel{L.R.}{\Longrightarrow} \lim_{x\to\infty} \left(\frac{1/x}{1}\right) = \lim_{x\to\infty} \left(\frac{1}{x}\right) = 0$$

Now we have:

$$\lim_{x\to\infty}\left(x\left(1-\frac{\ln(x)}{x}\right)\right)$$

This limit has the form $\infty \cdot 1$. This is a determinate form, and, therefore, $\lim_{x\to\infty}(x-\ln(x)) = \infty$

(e) $\lim_{x\to\infty}\left(x\ln\left(\frac{1}{x}\right)\right)$

**Explanation.** As $x$ approaches $\infty$, $\frac{1}{x}$ approaches 0 from the right. So

$$\lim_{x\to\infty}\ln\left(\frac{1}{x}\right) = -\infty$$

Therefore, the limit in question is of the form $\infty \cdot -\infty$, which is a determinate form.
Thus,

$$\lim_{x\to\infty}\left(x\ln\left(\frac{1}{x}\right)\right) = -\infty$$

(f) $\lim_{x\to 0^+}(\sin(x)\cot(x))$

**Explanation.** Since $\lim_{x\to 0^+}\cot(x) = \infty$, this limit is of the form $0\cdot\infty$. This is an indeterminate form.
Note: $\cot(x) = \frac{\cos(x)}{\sin(x)}$. So

$$\lim_{x\to 0^+}(\sin(x)\cot(x)) = \lim_{x\to 0^+}\cos(x) = 1.$$